

HWS

何衍宇 HE Yanning 12311512

1. A Si p-n junction has donor and acceptor doping concentrations  $N_d$  and  $N_a$  of  $10^{16} \text{ cm}^{-3}$  and  $10^{18} \text{ cm}^{-3}$ , respectively. Assume  $kT = 0.026 \text{ eV}$  at  $300 \text{ K}$ ,  $n_i = 1.5 \times 10^{10} \text{ cm}^{-3}$  and  $\epsilon_r = 11.8$ . Given that the junction has an area of  $10^{-4} \text{ cm}^2$ , calculate the junction capacitance at (i) zero bias (ii) FB of  $0.65 \text{ V}$  and (iii) RB of  $2 \text{ V}$ .

(3.18 pF, 7 pF, 1.7 pF)

$$N_D = 10^{16} \text{ cm}^{-3}, N_A = 10^{18} \text{ cm}^{-3}$$

$$kT = 0.026 \text{ eV} \Leftarrow T = 300 \text{ K}$$

$$n_i = 1.5 \times 10^{10} \text{ cm}^{-3}, \epsilon_r = 11.8, A = 10^{-4} \text{ cm}^2$$

$$\Rightarrow \epsilon = \epsilon_r \cdot \epsilon_0 = 1.045 \times 10^{-12} \text{ F/cm}$$

$$W = \sqrt{\frac{2\epsilon}{q} \cdot \frac{N_A + N_D}{N_A N_D} \cdot (V_{bi} - V)}$$

$$C = \frac{\epsilon A}{W}, V_{bi} = \frac{kT}{q} \ln \frac{N_A N_D}{n_i^2}$$

a) zero bias

$$V_{bi} = \frac{kT}{q} \ln \frac{N_A N_D}{n_i^2} = (0.026 \text{ V}) \cdot \ln \frac{10^{16} \cdot 10^{18}}{(1.5 \times 10^{10})^2} = 0.8171 \text{ V}$$

$$W_0 = \sqrt{\frac{2\epsilon}{q} \cdot \frac{N_A + N_D}{N_A N_D} \cdot (V_{bi} - 0)}$$

$$= \left[ \frac{2 \cdot 1.045 \times 10^{-12} \text{ F/cm}}{1.6 \times 10^{-19} \text{ C}} \cdot \frac{10^{18} + 10^{16}}{10^{18} \cdot 10^{16}} \cdot 0.8171 \text{ V} \right]^{\frac{1}{2}}$$

$$= 3.283 \times 10^{-5} \text{ cm}$$

$$C = \frac{\epsilon \cdot A}{W} = \frac{1.045 \times 10^{-12} \text{ F/cm} \cdot 10^{-4} \text{ cm}^2}{3.283 \times 10^{-5} \text{ cm}} = 3.18 \text{ pF}$$

b) FB 0.65 V

$$\Rightarrow W = \sqrt{\frac{2\epsilon}{q} \cdot \frac{N_A + N_D}{N_A N_D} \cdot (V_{bi} - V)}$$

$$= \left[ \frac{2 \cdot 1.045 \times 10^{-12} \text{ F/cm}}{1.6 \times 10^{-19} \text{ C}} \cdot \frac{10^{18} + 10^{16}}{10^{16} \cdot 10^{18}} \cdot (0.8171 - 0.65) \right]^{\frac{1}{2}}$$

$$= 1.485 \times 10^{-5} \text{ cm}$$

$$C = \frac{\epsilon A}{W} = \frac{1.045 \times 10^{-12} \text{ F/cm} \cdot 10^{-4} \text{ cm}^2}{1.485 \times 10^{-5} \text{ cm}} = 7.03 \text{ pF}$$

c) RB 2V

$$W = \sqrt{\frac{2\epsilon}{q} \cdot \frac{N_A + N_D}{N_A N_D} \cdot (V_{bi} + V)}$$

$$= \left[ \frac{2 \cdot 1.045 \times 10^{-12} \text{ F/cm}}{1.6 \times 10^{-19} \text{ C}} \cdot \frac{10^{16} + 10^{18}}{10^{16} \cdot 10^{18}} \cdot (0.8171 + 2) \right]^{\frac{1}{2}}$$

$$= 6.096 \times 10^{-5} \text{ cm}$$

$$C = \frac{\epsilon \cdot A}{W} = \frac{1.045 \times 10^{-12} \text{ F/cm} \cdot 10^{-4} \text{ cm}^2}{6.096 \times 10^{-5} \text{ cm}} = 1.714 \text{ pF}$$

2. Room temperature (300 K) measurements on a silicon p<sup>+</sup>-n junction diode yield the following results: With a reverse bias of 4.3 V, the junction capacitance is 20 pF. When the reverse bias is changed to 0.55 V, the junction capacitance is 40 pF. If  $n_i = 1.5 \times 10^{10} \text{ cm}^{-3}$ ,  $\epsilon_r = 11.8$ , and the diode area is  $2 \times 10^{-3} \text{ cm}^2$ , determine the contact potential  $V_o$ , and the doping density of the n-side  $N_d$ .

(0.7 V,  $6 \times 10^{15} \text{ cm}^{-3}$ )

$$T = 300 \text{ K} \quad p^+ - n \quad RB: 4.3 \text{ V} \rightarrow V_1$$

$$C_1 = 20 \text{ pF}, \quad RB \rightarrow 0.55 \text{ V}, \quad C_2 = 40 \text{ pF}$$

$$n_i = 1.5 \times 10^{10} \text{ cm}^{-3} \quad V_2 \quad \epsilon_r = 11.8 \quad A = 2 \times 10^{-3} \text{ cm}^2$$

$$\epsilon = \epsilon_r \cdot \epsilon_0 = 1.045 \times 10^{-12} \text{ F/cm}$$

$$W = \sqrt{\frac{2\epsilon}{q} \cdot \frac{N_A + N_D}{N_A N_D} \cdot \sqrt{V_{bi} + V}} = \text{Const. 1} \cdot \sqrt{V_{bi} + V}$$

$$C = \frac{\epsilon A}{W} = \frac{\text{Const. 2}}{W} = \frac{\text{Const. 2}}{\text{Const. 1} \cdot \sqrt{V_{bi} + V}}$$

$$V_{bi} = \frac{kT}{q} \cdot \ln \frac{N_A N_D}{n_i^2}$$

$$\frac{C_2}{C_1} = \frac{400 \text{ pF}}{200 \text{ pF}} = \sqrt{\frac{V_{bi} + V_1}{V_{bi} + V_2}} = 2 \Rightarrow \frac{V_{bi} + 4.3}{V_{bi} + 0.55} = 4$$

$$\Rightarrow V_{bi} = 0.7 \text{ V}$$

$$(0.026 \text{ V}) \cdot \ln \frac{N_A N_D}{(1.5 \times 10^{10})^2} = 0.7 \text{ V}$$

Since one-side abrupt p<sup>+</sup>-n  $\Rightarrow N_A \gg N_D$

$$\Rightarrow \frac{N_A + N_D}{N_A N_D} \approx \frac{1}{N_D} \Rightarrow W = \sqrt{\frac{2\epsilon}{q} \frac{1}{N_D} \cdot \sqrt{V_{bi} + V_1}}$$

$$\Rightarrow C_1 = A \cdot \sqrt{\frac{\epsilon q}{2} \cdot \frac{N_D}{V_{bi} + V_1}}$$

$$= 2 \times 10^{-3} \text{ cm}^2 \cdot \sqrt{\frac{1.045 \times 10^{-12} \text{ F/cm} \cdot 1.6 \times 10^{-19} \text{ C}}{2}} \cdot \frac{N_D}{0.7 + 4.3}$$

$$= 20 \text{ pF}$$

$$\Rightarrow N_D = \left( \frac{20 \text{ pF}}{2 \times 10^{-3} \text{ cm}^2} \right)^2 \frac{10}{1.6 \times 10^{-19} \text{ C} \times 1.045 \times 10^{-12} \text{ F/cm}}$$

$$= 5.98 \times 10^{15} \text{ cm}^{-3}$$

$$\Rightarrow \text{contact potential } V_0 = V_{bi} = 0.7 \text{ V} ; N_D = 6 \times 10^{15} \text{ cm}^{-3}$$

3. Consider a silicon p<sup>+</sup>-n junction with  $N_d = 10^{16} \text{ cm}^{-3}$ . The area of the diode is  $10^{-4} \text{ cm}^2$ . Assume  $kT = 0.026 \text{ eV}$  at 300 K,  $n_i = 1.5 \times 10^{10} \text{ cm}^{-3}$  and  $\epsilon_r = 11.8$ . Holes in the n-region have a lifetime of  $0.5 \mu\text{s}$  with a diffusion coefficient of  $10 \text{ cm}^2/\text{s}$ . Calculate the storage capacitance of the p-n junction at a forward bias of  $0.65 \text{ V}$ .

$$\text{p}^+\text{-n} \quad N_D = 10^{16} \text{ cm}^{-3} \quad A = 10^{-4} \text{ cm}^2 \quad (2.23 \text{ nF})$$

$$T = 300 \text{ K} \Rightarrow kT = 0.026 \text{ eV}, \quad n_i = 1.5 \times 10^{10} \text{ cm}^{-3}, \quad \epsilon_r = 11.8$$

n life-time:  $0.5 \mu s$ ,  $D = 10 \text{ cm}^2/s \rightarrow FB: 0.65 \text{ V}$

$$\Rightarrow \tau_n = 0.5 \mu s, D_n = 10 \text{ cm}^2/s$$

$$\Rightarrow C_s = \frac{dQ}{dV} = \tau_n \cdot \frac{dI_F}{dV} = \tau_n \cdot \frac{I_F}{V_T} \rightarrow V_T = \frac{kT}{q}$$

$$\Rightarrow I_s = \frac{q \cdot A \cdot D_p \cdot n_i^2}{N_D \cdot L_p} \quad \text{since } D_p = D_n, L_p = \sqrt{D_p \tau_p}$$

$$\tau_p = \tau_n \quad \checkmark$$

$$\Rightarrow I_s = \frac{1.6 \times 10^{-19} \text{ C} \cdot 10^{-4} \text{ cm}^2 \cdot 10 \text{ cm}^2/s \cdot (1.5 \times 10^{10} \text{ cm}^{-3})^2}{10^{16} \text{ cm}^{-3} \cdot \sqrt{10 \text{ cm}^2/s \cdot 0.5 \times 10^{-6} \text{ s}}}$$

$$= 1.610 \times 10^{-15} \text{ A}$$

$$\Rightarrow I_F = I_s \cdot \exp\left(\frac{qV}{kT}\right)$$

$$= (1.610 \times 10^{-15}) \cdot \exp\left(\frac{0.65}{0.026}\right)$$

$$= 1.16 \times 10^{-4} \text{ A}$$

$$\Rightarrow C_s = (0.5 \times 10^{-6} \text{ s}) \cdot \frac{1.16 \times 10^{-4} \text{ A}}{0.026} = 2.23 \text{ nF}$$

4. An ideal Schottky contact is formed on n-type Si having a donor concentration of  $10^{17} \text{ cm}^{-3}$ . The bandgap of Si at 300 K is 1.1 eV. The metal work function  $\phi_m$  is 5.4 eV and the electron affinity  $\chi$  is 4.5 eV at 300K. Calculate the semiconductor work function  $\phi_s$ . Also draw the energy band diagrams (a) before and after contact formation at equilibrium (b) at a forward bias of 0.2 V and (c) reverse bias voltage of 3.2 V. Assume  $n_i = 1.5 \times 10^{10} \text{ cm}^{-3}$ .

$$N_D = 10^{17} \text{ cm}^{-3}, E_g = 1.1 \text{ eV}, \phi_m = 5.4 \text{ eV}, \chi = 4.5 \text{ eV} \quad (4.642 \text{ eV})$$

$$\phi_s = \chi + (E_c - E_f)/q$$

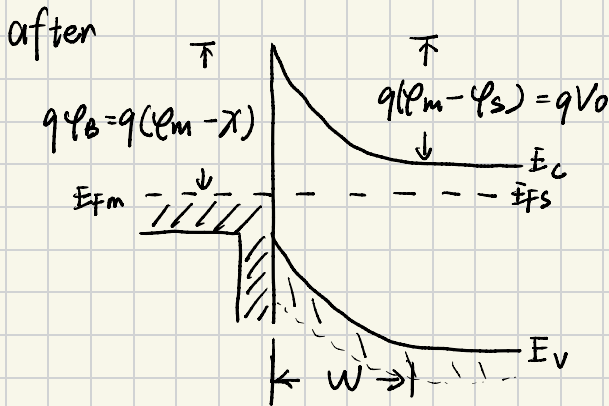
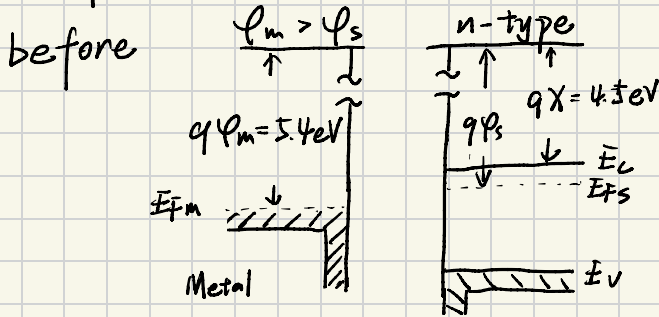
$$E_c - E_f = kT \cdot \ln \frac{N_c}{N_D} \quad N_c = 2.8 \times 10^{19}$$



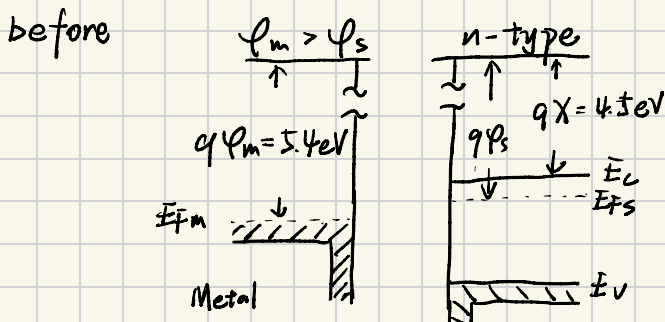
$$\Rightarrow q\varphi_s = q\chi + (E_c - E_f)$$

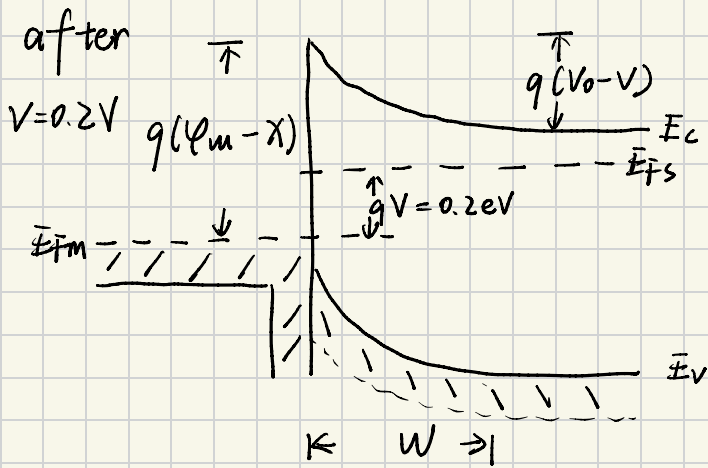
$$\Rightarrow q\varphi_s = 4.642 \text{ eV}$$

a) equilibrium



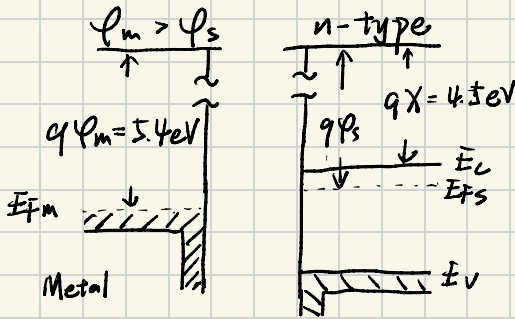
b) fb 0.2V





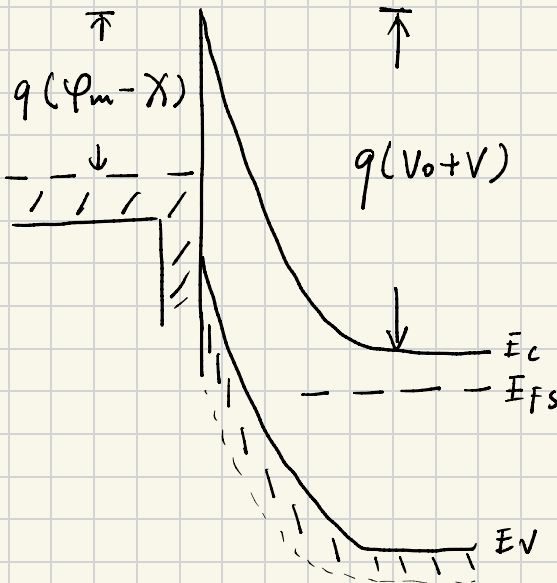
c) nb 3.2V

before

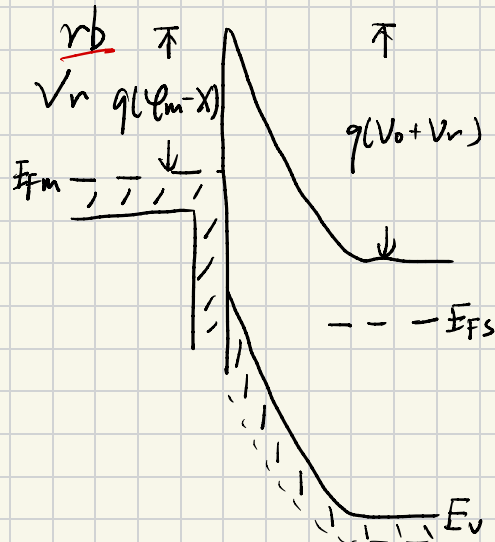
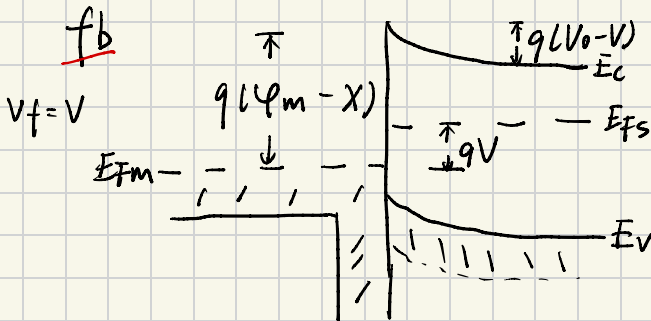
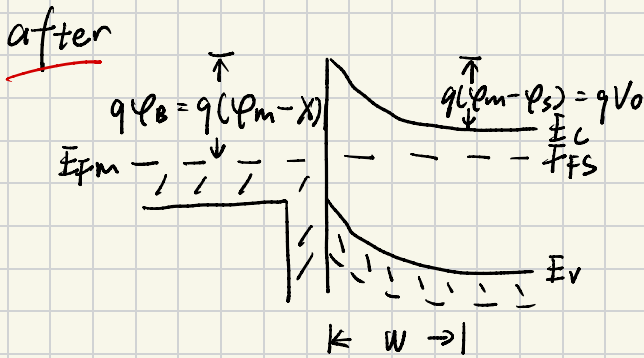
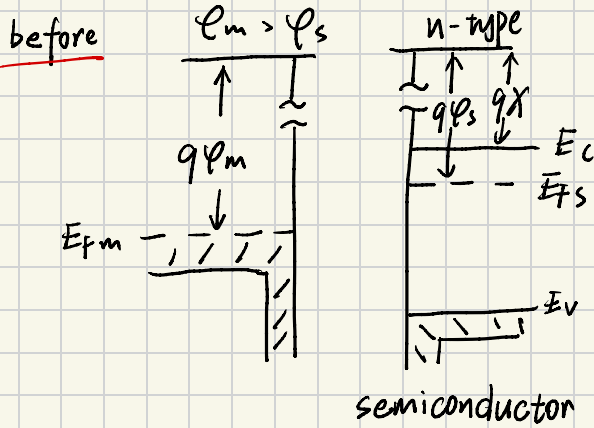


after

$V = 3.2V$



5. Draw the energy band diagrams of a Schottky contact before and after the formation of contact at the thermal equilibrium based on metal and n-type semiconductor. Repeat the energy band diagrams of the contact at forward and reverse bias, clearly showing the changes in the energy levels due to biases.



6. A metal semiconductor junction is formed between an n-doped Si and aluminum. It is given that the Si has a doping concentration  $N_d = 10^{17} \text{ cm}^{-3}$  and a work function

$\phi_s = 4.191 \text{ eV}$ , and aluminum has a work function  $\phi_m = 4.3 \text{ eV}$ . Assume that  $kT = 0.026 \text{ eV}$ , and Si has a bandgap of  $1.1 \text{ eV}$  and intrinsic concentration  $n_i = 1.5 \times 10^{10} \text{ cm}^{-3}$ . Deduce the electron affinity  $\chi$  of Si. Draw the energy band diagrams before and after the junction formation under equilibrium condition. Hence determine whether the junction formed is rectifying or ohmic. All the relevant energies must be clearly indicated in the band diagrams.

(4.05 eV, Schottky contact)

$$N_D = 10^{17} \text{ cm}^{-3}, \phi_s = 4.191 \text{ eV}, \phi_m = 4.3 \text{ eV}$$

$$kT = 0.026 \text{ eV}, E_g = 1.1 \text{ eV}, n_i = 1.5 \times 10^{10} \text{ cm}^{-3}$$

$$\Rightarrow q\chi$$

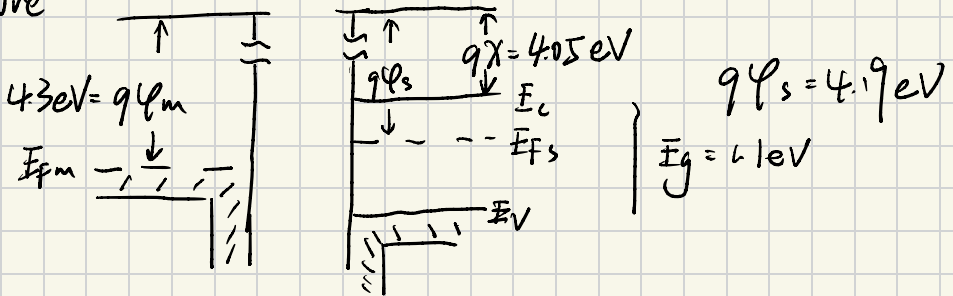
$$E_c - E_f = kT \cdot \ln \frac{N_c}{N_D} = 0.14650 \text{ eV}$$

$$q\phi_s = q\chi + (E_c - E_f)$$

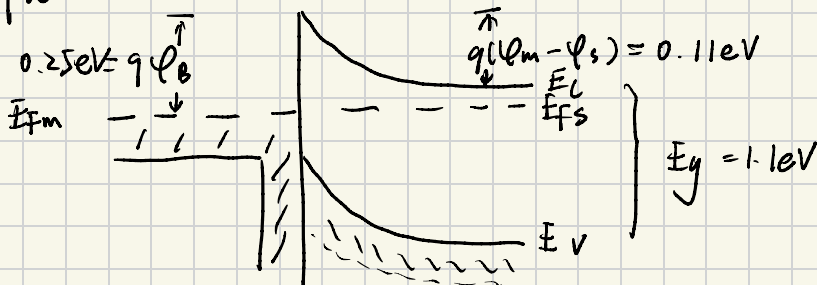
$$\Rightarrow q\chi = q\phi_s - (E_c - E_f) \approx 4.05 \text{ eV}$$

$$\Rightarrow q\phi_B = q(\phi_m - \chi) = 0.25 \text{ eV} \quad \phi_m > \phi_s$$

before



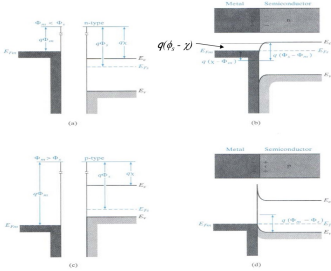
after



n-type  
since  $\phi_m > \phi_s \Rightarrow$  rectifying  $\Rightarrow$  Schottky contact

7. Discuss the operation of Ohmic contacts based on energy band diagrams.

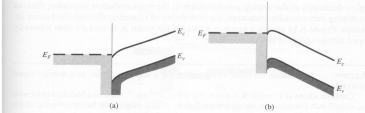
#### MS Contacts - Ohmic Contacts



Ohmic metal-semiconductor contacts: (a)  $\phi_m < \phi_s$  for an n-type semiconductor, and (b) the equilibrium band diagram for the junction; (c)  $\phi_m > \phi_s$  for a p-type semiconductor, and (d) the junction at equilibrium.

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#### MS Contacts - Ohmic Contacts



Ideal energy band diagram of a metal-n-semiconductor ohmic contact (a) with a positive voltage applied to the metal and (b) with a positive voltage applied to the semiconductor.

When a +ve voltage is applied to the metal, electrons can flow down-hill from the semiconductor into the metal.

When a +ve voltage is applied to semiconductor, electrons can easily flow over the (small) barrier from the metal into the semiconductor.

Hence this contact is called an ohmic contact.

$\Rightarrow$  n-type  $q\phi_m < q\phi_s$

$\Rightarrow$  net transfer of electrons

from metal to the semiconductor

$\Rightarrow$  p-type  $q\phi_m > q\phi_s$

$\Rightarrow$  Ohmic contact is formed when metal-semiconductor interface does not present a significant potential barrier to carrier flow

$\Rightarrow$  +ve voltage is applied to metal

electrons can flow down-hill from semiconductor to metal

$\Rightarrow$  +ve voltage is applied to semiconductor

electrons can easily flow over the small barrier

from metal into the semiconductor

$\Rightarrow$  Accumulation of majority carriers occur in these cases